Chunjiang Liu

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Electrical and Computer Engineering | GPA: 3.83/4.00 — 2023 - 2024
Courses: *Optimization, Computer Vision, Vision Learning and Recognition, Deep Generative Modeling*

University of Electronic Science and Technology of China (UESTC) — Chengdu, China
Bachelor of Communication Engineering | GPA: 3.84/4.00 — 2018 - 2022
Courses: *Calculus, Linear Algebra, Probability and Statistics, Signals and Systems*

TECHNICAL SKILLS

Languages & Frameworks: Python, Go, TS, SQL, PyTorch, FastAPI, HuggingFace
Data & Tools: PostgreSQL, MongoDB, LanceDB, Incus/LXD, Claude Agent SDK

EXPERIENCE

BAKE AI INC. — San Francisco, CA
Member of Technical Staff — 2026

- Fine-tuned Qwen3.5-35B-A3B (LoRA) into the aesthetic-judgment model behind ArtArena (+4-pt pass³, approaching a 397B model), trained on an automated 3D-rendering and text-to-image data pipeline.
- Engineered Eval, an agentic-RL evaluation harness running models across 600+ verifier environments with reward-function scoring, LLM-judge grading, and real-time rollout and trajectory inspection.
- Architected DatasetVecDB, a multimodal data-curation pipeline embedding text/image/video/audio (Jina-v3, EVA02-CLIP, BEATs) into a vector store for foundation-model training-data discovery.
- Built Agent Sandbox, isolated multi-cluster execution (Incus/LXD VMs + containers, Go orchestrator over Tailscale) hosting RL environments and agent rollouts with full lifecycle control and monitoring.

CARNEGIE MELLON UNIVERSITY, ECE — Pittsburgh, PA
Research Associate | Laszlo A. Jeni — 2024 - 2025

- Built MOSIV, a Gaussian framework recovering continuous per-object physics from multi-object video, beating OmniPhysGS (~30× lower Chamfer) and the stronger CoupNeRF (~2×, +8 dB future PSNR).
- Contributed to MASIV, a material-agnostic framework with learnable neural constitutive models (no material priors), outperforming PAC-NeRF, Spring-Gaus, and NeuMA on standard benchmarks.
- Contributed to Total-Editing, a unified head-avatar framework controlling appearance, motion, and lighting via an MLS deformation field, unifying face reenactment and portrait relighting end-to-end.

PROJECTS

- **Simulable Worlds from Text** - Plausible renders can hide fake physics/dynamics; introduced plan-grounded decomposition, a verifiable scene-graph protocol (object grounding, structural contact), and an engine-verified execute-inspect-repair loop for simulation-ready assets (MPR 0.830 vs. VIGA 0.674).

- **RAM-Bench** - Agents lose state across context resets, breaking continuation and debugging; framed memory as conditional compression and introduced result replay, where a fresh agent reconstructs prior artifacts and state from memory alone (350 tasks; fidelity 0.718 to 0.884, or 0.974 with replay-guided writing).

PUBLICATION

- **Chunjiang Liu**, X. Wang, H. Chen, Y. Zhao, Ming-Hsuan Yang, Laszlo A. Jeni, "Simulable Worlds from Text: LLM Agents for 4D Scene Creation", NeurIPS 2026 (under review)
- Y. Feng*, Y. Li*, **Chunjiang Liu***, Y. Chen, F. Jiang, Y. Huang, H. Hua, Z. Yuan, K. Zheng, L. Niu, B. Ramasubramanian, Basel Alomair, Xiangliang Zhang, Misha Sra, Z. Chen, Radha Poovendran, Z. Xu, "Visual Aesthetic Benchmark: Can Frontier Models Judge Beauty?", *COLM 2026* (*Equal contribution)
- **Chunjiang Liu**, X. Wang, Q. Lin, A. Xiao, H. Chen, S. Wen, H. Zhang, L. Qi, Ming-Hsuan Yang, Min Xu, Laszlo A. Jeni, Y. Zhao, "Multi-Object System Identification from Videos", *ICLR 2026*
- Y. Zhao, H. Chen, **Chunjiang Liu**, Z. Li, Charles Herrmann, Junhwa Hur, Y. Li, Ming-Hsuan Yang, Bhiksha Raj, Min Xu, "Toward Material-Agnostic System Identification from Videos", *ICCV 2025*
- Y. Zhao, **Chunjiang Liu**, H. Chen, Bhiksha Raj, Min Xu, Tadas Baltrusaitis, M. Rundle, H. Wu, K. Ghasedi, "Total-Editing: Head Avatar with Editable Appearance, Motion, and Lighting", *ICCV* 2025, *E2E3D*